Jiawei Zhang

Jiawei.Zhang@uky.edu | jiaweizhang.site

Research Interests

Machine Learning, privacy-aware decentralized learning, deep learning, statistical inference, and model selection and diagnostics.

Employment

Dr. Bing Zhang Department of Statistics , University of Kentucky, Lexington

Assistant Professor in Statistics *2023-*

Education

School of Statistics, University of Minnesota, Twin Cities

Ph.D. in Statistics *2017 - 2023*

- Ph.D. candidate advised by Prof. Yuhong Yang and Prof. Jie Ding

School of Statistics, University of Minnesota, Twin Cities

M.S. in Statistics *2015–2017*

School of Statistics and Management, Shanghai University of Finance and Economics, Shanghai

Bachelor of Economics *2011–2015*

Research Projects

Foundations of Machine Learning Evaluation [2, ??]

How to assess whether a trained model is sufficiently accurate?

- Developed a methodology from a nonparametric testing perspective to assess modern classification methods such as tree ensembles and neural networks

Foundations of Model Selection [3]

How to identify the most appropriate model for tasks in new data domains?

- Developed a cross-validation method for statistically reproducible comparison of multiple models or hyperparameters based on adaptive data-splitting

Decentralized Multi-Party Learning [1, 4]

How to facilitate learning entities (e.g., government agencies, research institutions, companies) with various data modalities to assist each other?

- Developed an *Assisted Learning* methodology that does not require sharing datasets or model parameters to autonomously assist each other
- Developed a multi-party learning mechanism in which a learning entity that receives assistance is incentivized to assist others, thus promoting mutual benefits

AI-enabled Cyber-Physical-Biological System for Organoid Maturation [6]

How to provide a reliable quantification of the maturation stage of man-made cardiac tissues using multimodal measurements at a single-cell level?

- Collaborated with bioengineering scientists to develop a multi-task learning system for nano-level sensing that substantially improved the state-of-the-art

In-Depth Review of Information Criteria [5]

When and how to use information criteria appropriately?

- Completed a review of information criteria that focuses on the fundamental challenges in model selection, insights into the theoretical properties of information criteria and their connections with other related methods, recent research advancements, and clarification of misleading folklore and practical guidelines

Teaching

Introduction to Data Science STA305 — *University of Kentucky*

2025

- Designed and taught this new undergraduate course for Statistics majors. Topics included data visualization, wrangling, and reproducible workflows (R, R Markdown, Git/GitHub); data ethics (privacy, bias, and misrepresentation); and statistical modeling and inference using a simulation-based approach. Incorporated modules on web scraping, tidy data principles, and model validation, with optional extensions to Shiny apps, text analysis, and Bayesian methods.

Introduction to Probability STA320 — *University of Kentucky*

2024

- A class for undergraduate students. It covers set theory; fundamental concepts of probability, including conditional and marginal probability; random variables and probability distributions (discrete and continuous); expected values and moments; moment-generating functions; random experiments; distributions of random variables and functions of random variables; limit theorems.

Introduction to Linear Models and Experimental Design STA603 — *University of Kentucky*

2024, 2025, 2026

- A core class in the M.S. Statistics program. It covers the multivariate normal distribution, linear models in matrix notation, multiple linear regression (distributional results, categorical predictors, interactions, connection to ANOVA, sums of squares, diagnostics), and professional presentation of results.

Introduction to Statistical Methods STA602 — *University of Kentucky*

2023, 2025

- A core class in the M.S. Statistics program. It introduces concepts and techniques for data analytical skills. The topics include sampling distributions, statistical models, point estimates and confidence intervals, and significance testing.

Introduction to Probability and Statistics STAT3021 (graduate instructor) — *University of Minnesota*

2020

- Covered basic probability theory, random variables, sampling distributions, and statistical inference for 139 undergraduate students.

Applied Regression Analysis STAT5302 (teaching assistant) — *University of Minnesota*

2017-2018

- The classes consisted of around 40 students, including undergraduate students from diverse disciplines and master or Ph.D. students from, e.g., computer science and business school. Covered estimation, testing, and prediction of regression models.

Software

BAGofT

- Author and maintainer of the R package *BAGofT* published in CRAN that implements the methods in [2] for assessing the goodness-of-fit of binary classifiers

Publications

Papers published (peer reviewed)

1. **J. Zhang**, J. Ding, Y. Yang, "Additive-Effect Assisted Learning," Journal of the Royal Statistical Society Series B: Statistical Methodology, (2025)
2. **J. Zhang**, J. Ding, Y. Yang, "Is a Classification Procedure Good Enough? A Goodness-of-Fit Assessment Tool for Classification Learning," Journal of the American Statistical Association, (2023)
3. **J. Zhang**, J. Ding, Y. Yang, "Targeted Cross-Validation," Bernoulli, (2023)
4. X. Wang, **J. Zhang**, M. Hong, Y. Yang, J. Ding, "Parallel Assisted Learning," IEEE Transactions on Signal Processing, (2022)
5. **J. Zhang**, Y. Yang, J. Ding, "Information Criteria for Model Selection," Wiley Interdisciplinary Reviews: Computational Statistics, (2023)
6. X. Tang, **J. Zhang** (co-first author), Y. He, X. Zhang, Z.Lin, S. Partarrieu, E. B. Hanna, Z. Ren, Y. Yang, X. Wang, N. Li, J. Ding, J. Liu, "Multi-task learning for single-cell multi-modality biology," Nature Communications, (2023)
7. E. Diao, G. Wang, **J. Zhang**, Y. Yang, J. Ding, V. Tarokh, "Pruning Deep Neural Networks from a Sparsity Perspective," International Conference on Learning Representations, (2023)

8. C. Chen, **J. Zhang** (co-first author), J. Ding, Y. Zhou, “Assisted Unsupervised Domain Adaptation,” IEEE International Symposium on Information Theory, (2023)

Manuscripts Under Review

9. Jin Du, Xinhe Zhang, Hao Shen, Xun Xian, Ganghua Wang, **Jiawei Zhang**, Yuhong Yang, Na Li, Jia Liu, and Jie Ding, “Conquering Catastrophic Forgetting in Machine Learning by Neural Representational Drift-inspired Networks.”
10. Zhecheng Sheng, **Jiawei Zhang**, Enmao Diao, “Toward Unifying Group Fairness Evaluation from a Sparsity Perspective.”

Manuscripts Ready

11. Zeya Wang, **Jiawei Zhang**, “A Nearest-neighbor Perspective on Internal Clustering Validation.”
12. Chenglong Ye, **Jiawei Zhang**, Yuhong Yang, “Cross Validation Importance Learning (CVIL).”

Presentations

Is a Classification Procedure Good Enough?—A Goodness-of-Fit Assessment Tool for Classification Learning *2024*

- University of Louisville Department of Bioinformatics & Biostatistics Seminar

Additive-Effect Assisted Learning *2024*

- 7th International Conference on Econometrics and Statistics (EcoSta 2024)

Is a classification procedure good enough?-A goodness-of-fit assessment tool for classification learning *2024*

- Southwestern University of Finance and Economics Department of Statistics Seminar

Is a classification procedure good enough?-A goodness-of-fit assessment tool for classification learning *2024*

- 2024 Annual ASA Florida Chapter Meeting

Is a classification procedure good enough?-A goodness-of-fit assessment tool for classification learning *2023*

- University of Kentucky Applied Math Seminar

Targeted Cross-Validation (poster presentation) *2019*

- Cook’s Distance and Beyond: A Conference Celebrating the Contributions of R. Dennis Cook

Honors

2022 **Martin Statistics Travel Award**, School of Statistics, University of Minnesota
2020 **Martin-Buehler Fellowship in Statistics**, School of Statistics, University of Minnesota
2018 **Martin-Buehler Fellowship in Statistics**, School of Statistics, University of Minnesota

Services

Reviewing for Journals (Dates indicate when the review requests were accepted, and each item in this list corresponds to a different paper.)

- IEEE Transactions on Signal Processing (2022/07, 2022/12, 2023/03)
- Transactions on Signal Processing (2023/04)
- Annals of Statistics (2023/07)
- Journal of Computational and Graphical Statistics (2023/08)
- Journal of Computational and Graphical Statistics (2023/12)
- Econometric Reviews (2024/04)
- Journal of Applied Statistics (2024/04)
- Journal of the American Statistical Association (2024/09, 2025/03)
- Discover Artificial Intelligence (2025/03)
- IEEE Transactions on Information Theory (2025/07)
- American Journal of Mathematical and Management Sciences (2026/01)
- IEEE Transactions on Information Theory (2026/02)
- IEEE Transactions on Information Theory (2026/03)
- Journal of the American Statistical Association (2026/04)
- Journal of the American Statistical Association (2026/04)

Master of Applied Statistics (MAS) Program Capstone Committee *2026*

- Served as an examiner for MAS capstone projects.

Outside Examiner for Doctoral Final Examination *2025*

- Served as an external examiner for a PhD candidate in Materials Science and Engineering in Fall 2025.

Committee Member of UK AI Machine Learning Hub *2023-2025*

- The UK Hub for AI/ML aims at transforming the educational and research capacity of AI/ML work at UK by building a centralized hub connecting AI/ML method consumers, users, and developers.
- Co-organized a machine learning symposium for researchers and a workshop for students
- Attended monthly meetings and communicated with researchers with diverse backgrounds

Committee Member of Departmental Comprehensive Examinations for Master Students *2023-2025*

- Served as a committee member for the applied comprehensive exams for master students in statistics. The responsibilities are exam design, grading, and determining the exam outcomes.

Mentor of K-12 Outreach Program *2019*

- Served as the mentor of 2022 AEOP High School Apprenticeship K-12 Outreach Program to enhance diversity, equity, and inclusion (sponsored by Army Research Office)
- Guided four high school students with diverse backgrounds from Minnesota, Missouri, and Virginia on hands-on AI projects

Program Committee Member of IEEE BigData Workshop *2021*

- Co-organized the 2021 IEEE BigData Workshop on Scalable Reinforcement Learning with Big Data

Student Representative for School of Statistics Faculty Search *2021*

- Organized the meetings between faculty candidates and graduate students, facilitated the evaluation of candidates, and presented the results to the faculty

School of Statistics Advisory Committee Meeting *2017.2018*

- Volunteered in the advisory committee meeting, communicated with influential statistical leaders outside the university, and improved the recognition of the school